2023-2024

S5 MATH CP

PAPER II

MC

Name: _____

Class: _____( )

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Five Final Examination

MATHEMATICS Compulsory Part
PAPER 2

03/06/2024

11:05 am – 12:20 pm (1¼ hours)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first write your Name Class and Class Number and insert the information required in the spaces provided. No extra time will be given for writing your Name, Class and Class number after the 'Time is up' announcement.

2. When told to open this book, you should check that all questions are there. Look for the words ' **END OF PAPER**' after the last question.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answer clearly; otherwise you will lose marks if the answers cannot be captured.

5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

6. No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.

The diagrams in this paper are not necessarily drawn to scale.

Choose the best answer for each question.

Section A

1. $9^{x-5} \cdot 5^{2x-10} =$

   A. 15^{2x-10}.

   B. 15^{3x-15}.

   C. 45^{2x-10}.

   D. 45^{3x-15}.

2. $4a^2 - 7ab + 3b^2 - 4a + 3b =$

   A. $(4a - 3b)(a - b - 1)$.

   B. $(4a - 3b)(a + b + 1)$.

   C. $(4a + 3b)(a - b - 1)$.

   D. $(4a + 3b)(a + b - 1)$.

3. If $\frac{2x}{x-3} = \frac{3y}{y+2}$, then $x =$

   A. $-\frac{9y}{y+4}$.

   B. $\frac{9y}{y-4}$.

   C. $\frac{9y}{4-y}$.

   D. $\frac{9y}{y+4}$.

4. If $m - 3n + 5 = 4m - 9n + 1 = 6$, then $n =$

   A. -2.

   B. $-\frac{1}{3}$.

   C. $\frac{1}{3}$.

   D. 2.

5. A sum of \$95 200 is deposited at an interest rate of 6% per annum for 8 years, compounded half-yearly. Find the amount correct to the nearest dollar.

   A. \$103 088
   B. \$152 768
   C. \$153 484
   D. \$241 841

6. $\sqrt[4]{848}$ =

   A. 6 (correct to the nearest integer).
   B. 5.39 (correct to 2 decimal places).
   C. 5.396 (correct to 3 significant figures).
   D. 5.3963 (correct to 4 decimal places).

7. A point V is translated 6 units downwards. Then the image is rotated anticlockwise about the origin through 180° and reflected about the x-axis to $V_1(-4, -8)$. The coordinates of V are

   A. $(4, -8)$.
   B. $(-4, 8)$.
   C. $(4, -2)$.
   D. $(-4, 2)$.

8. The solution of $-2x < 18 < 3x$ is

   A. $x > -9$.
   B. $x > 0$.
   C. $x > 6$.
   D. $-9 < x < 6$.

9. Let p and q be constants. If $p(x + 1)(2x - 3) + q(x + 2)^2 \equiv -3x^2 + 6x + 10$, then p =

   A. –2.
   B. –1.
   C. 1.
   D. 2.

10. If x, y and z are non-zero numbers such that $3x = z$ and $2y = 5z$, then $(y - x) : (y + z)$ =

    A. 7 : 13.
    B. 13 : 7.
    C. 13 : 21.
    D. 21 : 13.

11. Let $f(x) = x^2 + kx - 14$, where k is a constant. If $f(2) = -18$, then $f(k) =$
    A. -14
    B. -4
    C. 14
    D. 18

12. Let $g(x) = 3x^3 - kx + 4$. If $g(x)$ is divisible by $x - 2$, find the remainder when $g(x)$ is divided by $x + 1$.
    A. -7
    B. 7
    C. 21
    D. 15

13. Which of the following statements about the graph of $y = -8 + (5 - x)^2$ is true?
    A. The graph opens downwards.
    B. The x-intercept of the graph is 5.
    C. The y-intercept of the graph is -8.
    D. The graph passes through the point $(5, -8)$.

14. The mean of 10 numbers is 24. If the mean of the first 6 numbers is 18, then the mean of the other 4 numbers is
    A. 21.
    B. 24.
    C. 30.
    D. 33.

15. It is given that $f(x)$ partly varies directly as x and partly varies directly as x^3. If $f(2) = 8$ and $f(3) = -33$, then $f(-1) =$
    A. -41.
    B. -13.
    C. 13.
    D. 41.

16. It is given that y varies inversely as x. If x is increased by 25%, then y is decreased by
    A. 20%.
    B. 25%.
    C. 80%.
    D. 125%.

17. If x varies directly as y and inversely as the cube of z, which of the following must be constant?

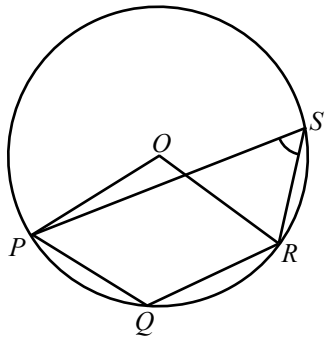
A. xyz^3

B. $\dfrac{z^3}{xy}$

C. $\dfrac{x}{yz^3}$

D. $\dfrac{y}{xz^3}$

18. In the figure, O is the centre of the circle $PQRS$.



If $OPQR$ is a rhombus, then $\angle PSR =$

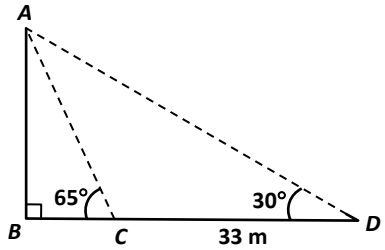
A. 55°.

B. 60°.

C. 65°.

D. 70°.

19.



In the figure, AB is a vertical flagpole. BCD is a straight line on the horizontal ground. The angles of elevation of A from C and D are 65° and 30° respectively. C and D are 33 m apart. Find the distance between B and C, correct to 3 significant figures.

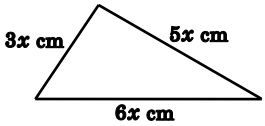
A. 12.2 m

B. 16.0 m

C. 26.1 m

D. 28.8 m

20.



The area of the triangle in the figure is

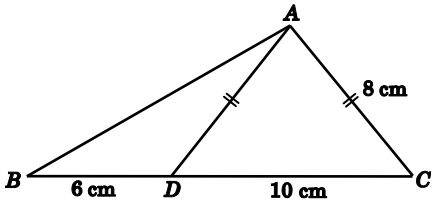
A. $\sqrt{8}x^2$ cm^2.

B. $\sqrt{28}x^2$ cm^2.

C. $\sqrt{56}x^2$ cm^2.

D. $\sqrt{630}x^2$ cm^2.

21.



In the figure, BDC is a straight line. If $AC = AD = 8$ cm, $BD = 6$ cm and $DC = 10$ cm, find the length of AB, correct to 3 significant figures.

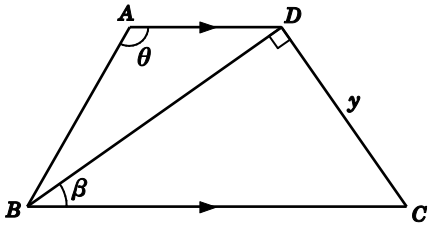
A. 6.32 cm

B. 6.93 cm

C. 11.0 cm

D. 12.6 cm

22.



In the figure, AB =

A. $\dfrac{y\cos\beta}{\sin\theta}$.

B. $\dfrac{y\tan\beta}{\sin\theta}$.

C. $\dfrac{y\sin\beta}{\sin\theta\tan\beta}$.

D. $\dfrac{\sin\beta}{y\sin\theta}$.

23. The rectangular coordinates of the point P are $(4, -4\sqrt{3})$. If P is rotated clockwise about the origin through 90°, then the polar coordinates of its image are
    A. $(8, 60^\circ)$.
    B. $(8, 150^\circ)$.
    C. $(8, 210^\circ)$.
    D. $(8, 240^\circ)$.

24. A point P moves such that it maintains an equal distance from a fixed point R. The locus of P is
    A. a straight line.
    B. a circle.
    C. a square.
    D. an ellipse.

25. A moving point P maintains an equal distance from $y = ax + b$ and $y = ax + 5b$, where a and b are non-zero real constants, then the locus of P is
    A. a straight line.
    B. a circle.
    C. a square
    D. a parabola.

26. The coordinates of the points A and B are $(-2, -1)$ and $(4, -9)$ respectively. Let P be a moving point in the rectangular coordinate plane such that $AP = BP$. Find the equation of the locus of P.
    A. $3x - 4y + 2 = 0$
    B. $3x - 4y - 23 = 0$
    C. $3x - 4y - 48 = 0$
    D. $4x + 3y + 11 = 0$

27. Two fair dice are thrown in a game. If the difference in the two numbers is less than 2, \$90 will be gained; otherwise, \$45 will be lost. Find the expected gain of the game.
    A. \$10
    B. \$15
    C. \$45
    D. \$60

28. Consider the circle C: $2x^2 + 2y^2 - 12x - 8y - 25 = 0$. Given that the coordinates of A is $(-2, 6)$. Suppose G is the centre of C and O is the origin. Which of the following is/are true?

I. $OG < 4$.

II. A lies outside C.

III. $\angle AOG$ is an acute angle.

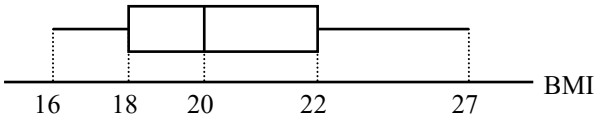
A. I and II only

B. I and III only

C. II and III only

D. I, II and III

29. The box-and-whisker diagram below shows the distribution of the body mass index (BMI) of a group of students. If the BMI exceeds 22, the student is overweight.



The percentage of overweight students is

A. 25%.

B. 50%.

C. 70%.

D. 75%.

30. $\{h - 5, h - 2, h, h + 2, h + 4, h + 7\}$ and $\{h - 8, h - 3, h + 1, h + 4, h + 1\}$ are two groups of numbers. Which of the following are true?

I. The two groups of numbers have the same mean.

II. The two groups of numbers have the same median.

III. The two groups of numbers have the same range.

A. II only

B. I and III only

C. II and III only

D. I, II and III

Section B

31. Convert $16^6 + 3 \times 16^4 + 22$ to a hexadecimal number.
    A. $A30016_{16}$
    B. $A30022_{16}$
    C. 1030016_{16}
    D. 1030022_{16}

32. For $0° \le \theta < 360°$, how many roots does the equation $4\cos^2\theta + 9\sin\theta - 6 = 0$ have?
    A. 5
    B. 4
    C. 3
    D. 2

33. Solve $\log_3(x + 4) + \log_3(x - 2) = 3$.
    A. $x = 5$ *or* -7
    B. $x = 3\sqrt{2} - 1$
    C. $x = 5$
    D. $x = 7$

34. It is given that $\log_2 y$ is a linear function of x. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are -6 and 6 respectively. If $y = \frac{m^x}{n}$, then
    A. $m = 1$ and $n = 16$.
    B. $m = 2$ and $n = 64$.
    C. $m = 4$ and $n = 64$.
    D. $m = 8$ and $n = -128$.

35. Let k be a constant. Find the range of values of k such that $2x^2 - (k + 5)x + k + 3 \ge 0$ for any real number x.
    A. All real numbers
    B. $k < -1$ or $k > -1$
    C. $k > -1$
    D. $k = -1$

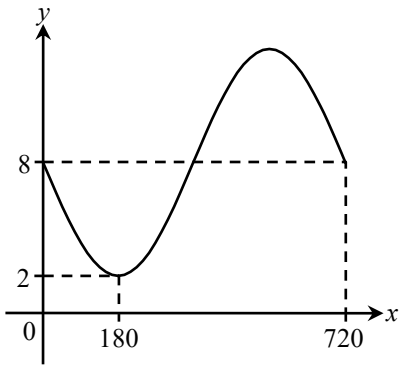
36. The figure shows

A. the graph of $y = 8 - 6\sin 2x°$.

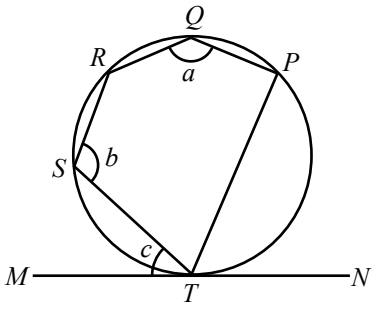
B. the graph of $y = 8 - 6\sin \frac{x°}{2}$.

C. the graph of $y = 2 - 6\sin 2x°$.

D. the graph of $y = 2 - 6\sin \frac{x°}{2}$.



37. In the figure, P, Q, R, S and T are points lying on the circle. MN is the tangent to the circle at T.



If TS bisects $\angle MTP$, which of the following must be true?

A. $a + b + 2c = 360°$

B. $a + b - 2c = 360°$

C. $a + 2c - b = 180°$

D. $b + 2c - a = 180°$

38. If the straight line $x - y + k = 0$ and the circle $x^2 + y^2 + 4x - 8y - 5 = 0$ intersect at P and Q, then the x-coordinate of the mid-point of PQ is

A. $-k + 2$.

B. $k - 2$.

C. $\frac{-k + 2}{2}$.

D. $\frac{k - 2}{2}$.

39. Find the range of values of k such that the straight line $y = x - 1$ and the circle $x^2 + y^2 + 3x - 5y + k = 0$ do not intersect with each other.

A. $k > -4$

B. $k < -4$

C. $k > 4$

D. $k < 4$

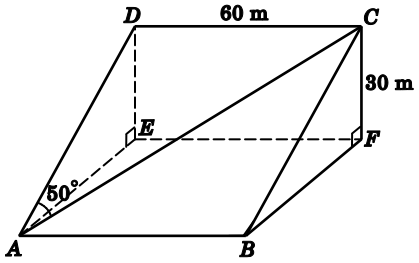
40. There are 9 boys and 11 girls in a class. 5 students are selected from the class to participate in a performance. Find the number of ways of selecting the students if at least 1 girl is selected.

A. 15 042
B. 15 378
C. 15 504
D. 15 630

41. 6 boys and 2 girls line up in a row at random. Find the probability that the 2 girls stand next to each other.

A. $\frac{1}{16}$
B. $\frac{1}{8}$
C. $\frac{1}{6}$
D. $\frac{1}{4}$

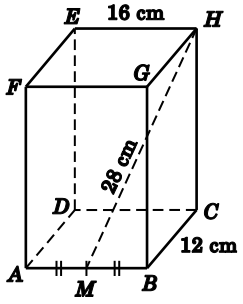
42.



In the figure, *ABCD* is a rectangular inclined plane and *ABFE* is the horizontal plane. *E* and *F* are the projections of *D* and *C* on the horizontal plane *ABFE* respectively. *ABFE* and *DEFC* are rectangles. It is given that $CF = 30$ m, $CD = 60$ m and $\angle CAD = 50°$. Find the inclination of the plane *ABCD*, correct to the nearest degree.

A. 23°
B. 25°
C. 30°
D. 37°

43.



In the figure, *ABCDEFGH* is a cuboid and *M* is the mid-point of *AB*. *EH* = 16 cm, *BC* = 12 cm and *HM* = 28 cm. Find the angle between the line *HM* and the plane *ABCD*, correct to the nearest degree.

A. 25°
B. 44°
C. 59°
D. 63°

44. In an examination, Peter gets 72 marks and his standard score is 1.2 while Wendy gets 57 marks and her standard score is −0.3. Find the standard deviation of the examination score.

A. 5 marks
B. 10 marks
C. 30 marks
D. 60 marks

45. The mode, the inter-quartile range and the variance of a group of numbers are 13, 22 and 8 respectively. If 5 is added to each number of the group and each resulting number is then divided by 2 to form a new group of numbers, find the mode, the inter-quartile range and the variance of the new group of numbers.

	Mode	Inter-quartile range	Variance
A.	9	11	2
B.	9	11	4
C.	9	22	2
D.	13	22	4

END OF PAPER